Install Wt before building ITSV

Choose Linux packages or download and build the C++ Web Toolkit from source.

OFFICIAL WT WEBSITE - DOWNLOAD HERE

https://www.webtoolkit.eu/wt/download

Wt 4.x is the C++ Web Toolkit. ITSV requires its headers and the **wt** and **wthttp** libraries. Wt must be installed separately.

OPTION A: LINUX PACKAGES (IF YOUR DISTRIBUTION PROVIDES THEM)

```
sudo apt update
sudo apt install build-essential libboost-all-dev curl \
    libssl-dev zlib1g-dev libwt-dev libwthttp-dev
```

For Debian/Ubuntu-based systems. Package names and availability vary; if these packages are unavailable, use Option B.

OPTION B: DOWNLOAD WT SOURCE AND COMPILE IT (DEBIAN/UBUNTU)

```
sudo apt update && sudo apt install build-essential cmake git curl \
    libboost-all-dev libssl-dev zlib1g-dev
git clone --depth 1 --branch 4.14.4 https://github.com/emweb/wt.git
cmake -S wt -B wt/build -DCMAKE_BUILD_TYPE=Release \
    -DCMAKE_INSTALL_PREFIX=/usr/local -DBUILD_EXAMPLES=OFF
cmake --build wt/build -j4
sudo cmake --install wt/build
sudo ldconfig
```

Wt 4.14.4 is available as source. It normally installs under `/usr/local` (ITSV default). With distro packages in `/usr`, use `make WT_PREFIX=/usr` on page 2.

ITSV SOURCE (GITHUB) https://github.com/akimov-seismology/itsv-wt

Build and run ITSV

With Wt installed (see page 1), build and run the local waveform viewer.

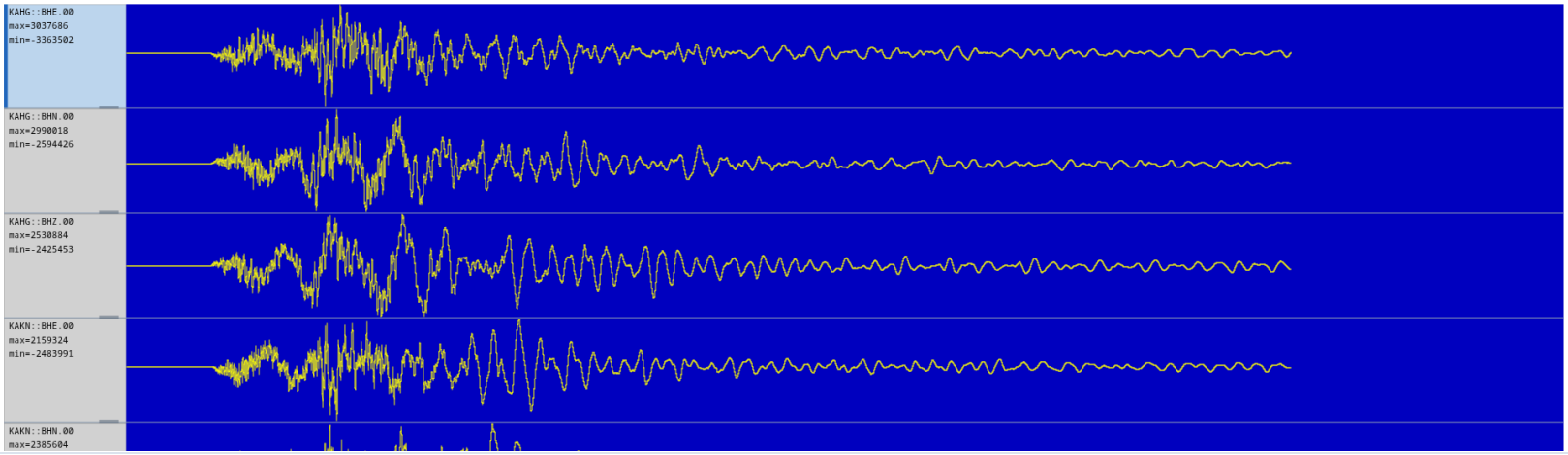
BEFORE YOU START

Linux with a C++17 compiler, GNU Make, **Wt 4.x** development libraries (wt and wthttp), Boost, OpenSSL and zlib. Have **curl** or **wget** available and internet access for the first build.

TERMINAL / clone from the ITSV GitHub repository

```
git clone https://github.com/akimov-seismology/itsv-wt.git
cd itsv-wt
make -j4 WT_PREFIX=/usr
# Instead: make -j4       (if Wt is in /usr/local)
make test                  # optional checks
./run
```

Next: Keep the terminal running. In a web browser, open **http://127.0.0.1:8081/**. ITSV is bound to the local computer.

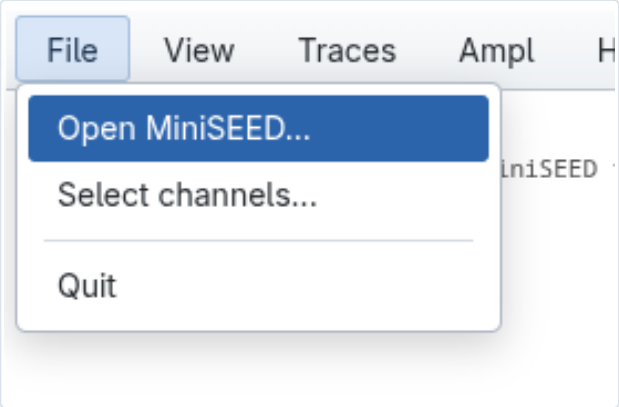


Example: six MiniSEED channels displayed in the main waveform pane.

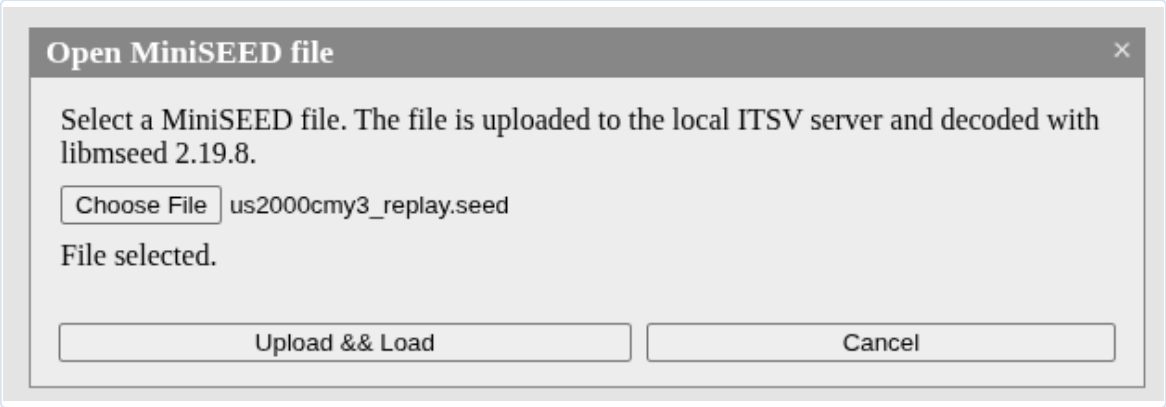
Open a MiniSEED recording

The file is selected in your browser and loaded by the ITSV server running on your computer.

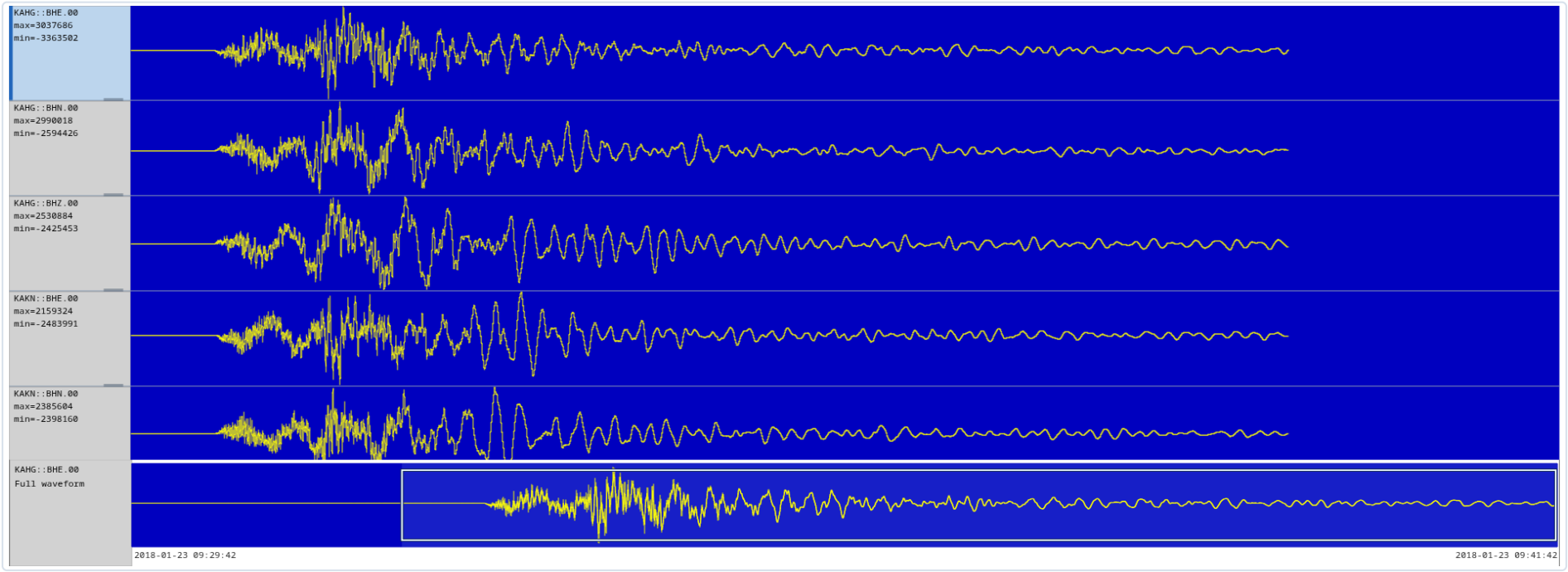
1 CHOOSE FILE > OPEN MINISEED...



2 CHOOSE A FILE, THEN UPLOAD & LOAD

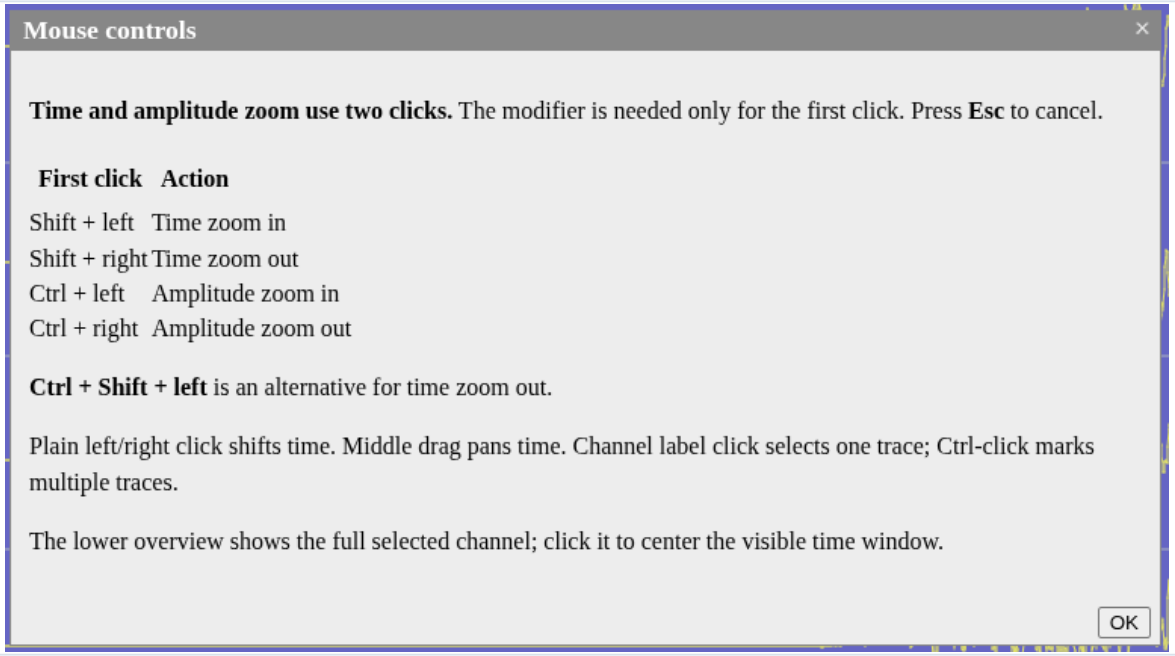


3 After loading, traces appear above, and the selected channel's full-waveform overview appears below.



Zoom, pan and use the overview

Most mouse zoom actions use two clicks: choose a start point and then an end point.



QUICK MOUSE CONTROLS

Shift + left: two-click time zoom in

Shift + right: time zoom out

Ctrl + left: amplitude zoom in

Ctrl + right: amplitude zoom out

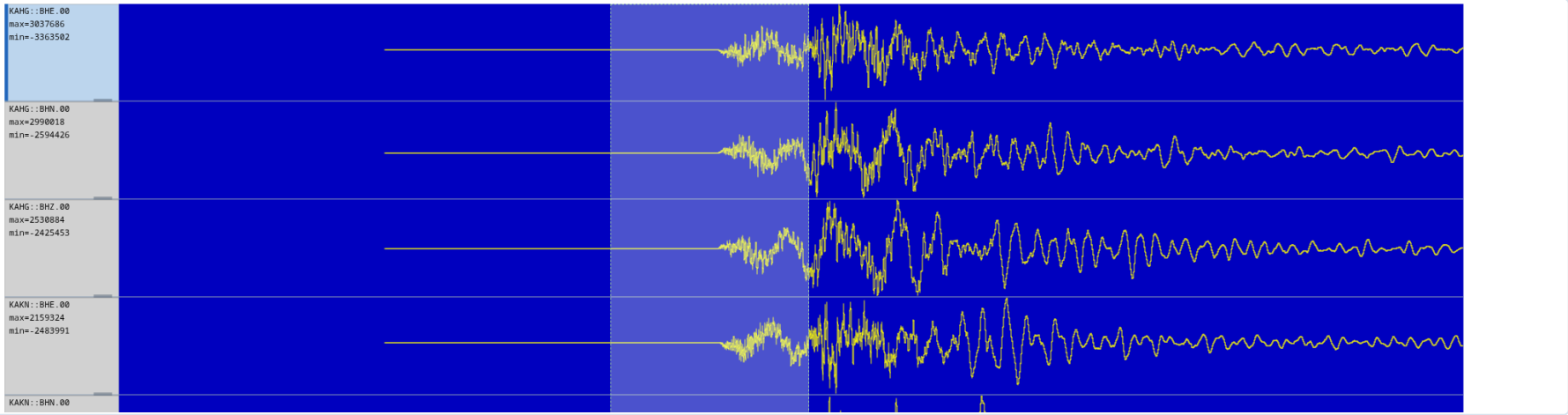
Middle drag: pan in time

Plain click: shift time

Esc: cancel a pending zoom

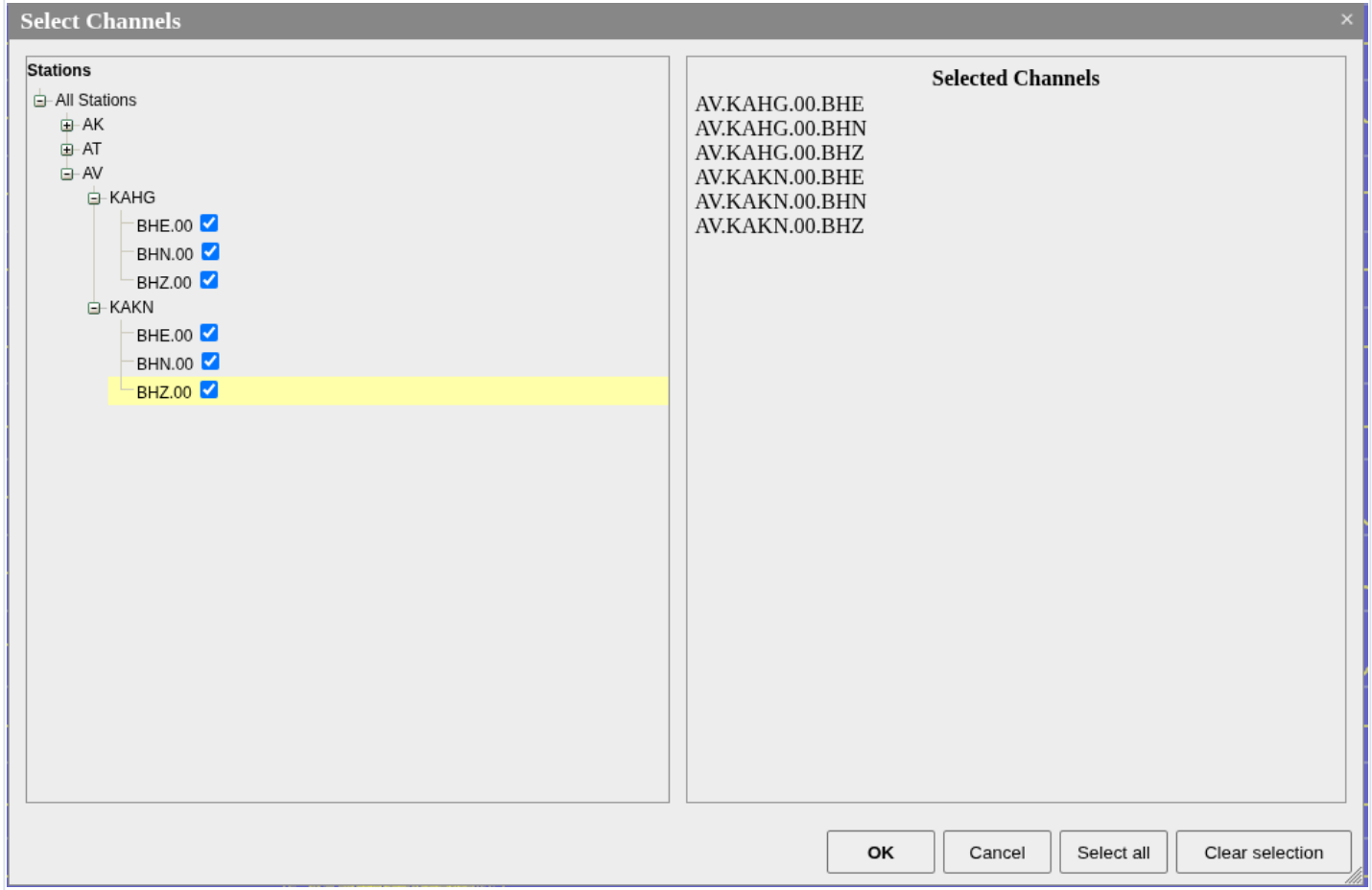
Click channel label: switch overview

Example: the translucent band marks a selected time interval during a two-click zoom.



Choose which channels to display

Use the channel tree when a MiniSEED file contains many stations and components.



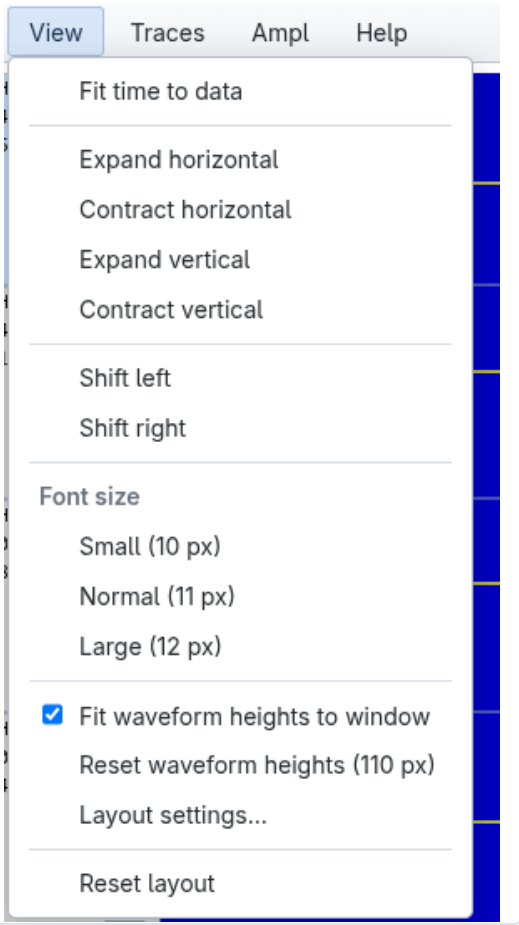
SELECT CHANNELS

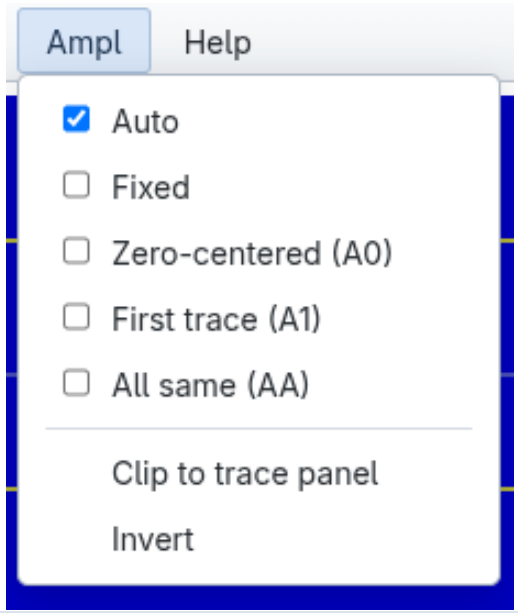
1. Choose **File > Select channels...**
2. Expand a network and station in the left tree.
3. Check channels you want to display.
4. Review the Selected Channels list on the right.
5. Select **OK** to update the waveforms.

Tip: Click a trace label to select one channel. Ctrl-click adds or removes it from the marked set. The **Traces** menu can keep, remove or restore traces.

Adjust waveform height and scaling

The View and Ampl menus control the layout without changing the MiniSEED recording.





VIEW: WAVEFORM SIZING

Fit waveform heights to window is on by default.

Each visible trace has an equal height. Minimum: **56 px**. More channels cause scrolling.

Drag any row divider to set a shared fixed height; select Fit to restore automatic sizing.

Layout settings... adjusts preview height, label width and text size.

AMPL: SCALING

Use **Auto**, **Fixed**, zero-centered, first-trace, or same-scale modes. A check mark shows the active mode.

IF SOMETHING DOES NOT WORK

Wt headers not found? Try the other build prefix: `make -j4` or `make -j4 WT_PREFIX=/usr`.

Wt resources not found? Launch with `WT_RESOURCES_DIR=/path/to/Wt/resources ./run`.

Blank viewer? Open a MiniSEED file first; confirm that `./run` is still running and that you are using port 8081.